$X_{24} = X_{14} - X_{13}$

$\vdots$

$X_{2n-1} = X_{1n-1} - X_{13}$

$X_{2n} = -X_{13}$

35	36	37	-- 3n
	46	47	-- 4n
		54	-- 57
			$\vdots$
			n-2 n

$X_{i-1\ i+1} = X_{i-2\ i+1} - X_{i-2\ i}$

$\vdots$

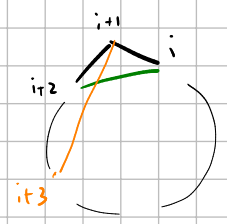
$X_{i-1,i+n-4} = X_{i-2,i+n-4} - X_{i-2,i}$

$X_{i-1,i+n-3} = -X_{i-2\ i}$

$X_{i-3,i-1}$

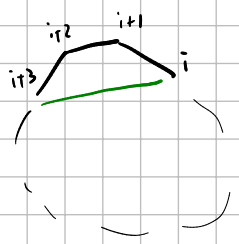
$X_{i\ i+2}$	$X_{i\ i+3}$	--	$X_{i\ i+n-3}$
	$X_{i+1\ i+3}$	--	$X_{i+1\ i+n-3}$
			$\vdots$
			$X_{i+n-5,\ i+n-3}$

$i = 1, \ldots, n$



crosses $(i, i+2)$ $(i+1\ i+3), (i+1\ i+4), \ldots, (i+1, i+n-1)$ [$1 + (n-3)$ things]

same as $i-1$

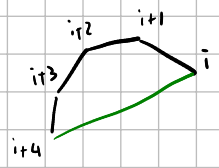


crosses $(i\ i+3)$

$(i+1\ i+4), (i+1\ i+5), \ldots, (i+1, i+n-1)$

$(i+2\ i+4), (i+2\ i+6) \ldots, (i+2, i+n-1)$

$$\left[\begin{array}{c} 1 + n - 4 + n - 4 \\ \parallel \\ 1 + 2(n-4) \end{array}\right]$$



crosses $(i\ i+4)$

$(i+1\ i+5), \ldots, (i+1, i+n-1)$

$\vdots$

$(i+3\ i+5), \ldots, (i+3, i+n-1)$

$$\left[\ 1 + 3(n-5)\ \right]$$

stop at $(i, i + \lfloor n/2 \rfloor)$

---

shift by -1

$X_{i-2\ i} = X_{i-3\ i} - X_{i-3\ i-1}$

$\vdots$

$X_{i-2,i+n-5} = X_{i-3,i+n-5} - X_{i-3,i-1}$

$X_{i-4,i-2} = -X_{i-3,i-1}$

$X_{i-1\ i+1}$	$X_{i-1\ i+2}$	--	$X_{i-1\ i+n-4}$
	$X_{i\ i+2}$	--	$X_{i\ i+n-4}$
			$\vdots$
			$X_{i+n-6,\ i+n-4}$

$i+n-6-(i-1)+1 = n-6+2 = n-4$

We can do a total of $n-4$ constraints where $i\ i+2$ is a free variable.

$$X_{24} = X_{14} - X_{13}$$
$$\vdots$$
$$X_{2n-1} = X_{1n-1} - X_{13}$$
$$X_{2n} = -X_{13}$$

$$\begin{array}{cccc} X_{35} & X_{36} & \cdots & X_{3n} \\ & X_{46} & & \vdots \\ & & \ddots & X_{n-2\,n} \end{array}$$

1

$$X_{13} = X_{n3} - X_{n2}$$
$$\vdots$$
$$X_{1n-2} = X_{nn-2} - X_{n2}$$
$$X_{1n-1} = -X_{n2}$$

$$\begin{array}{cccc} X_{24} & X_{25} & \cdots & X_{2n-1} \\ & X_{35} & & \vdots \\ & & \ddots & X_{n-3\,n-1} \end{array}$$

2

$3-n+s$

$$X_{68} = X_{58} - X_{57}$$
$$\vdots$$
$$X_{6\,n-3} = X_{5\,n+3} - X_{57}$$

$$\begin{array}{cccc} X_{7-n} & X_{8-n} & \cdots & X_{7-n\,s} \\ & \ddots & & \vdots \\ & & & X_{35} \end{array}$$

$n-4$

$n-2$ case

lean of $\dfrac{1}{X_{35}X_{46}}$ (things that are not 35 and 46)

$N = n-3$

$\dfrac{1}{X_{35}^{N}}$ all disappear

lean lim $n=4$ case

$$\frac{1}{X_{35}^{N-1}}\left(\sum_{(ij)\neq(35)} \frac{\cdots}{X_{ij}}\right)$$

$$\frac{1}{X_{35}^{N-2}}\left(\cdots \frac{\quad}{X_{i_1j_1}X_{i_2j_2}}\right)$$

X2n =-X13
X2n-1 = X1n-1 - X-13
...
X24 = X14 - X13

replace every 2 by 2 + j, and every 1, and ever 3 also by 1 and 3 + j

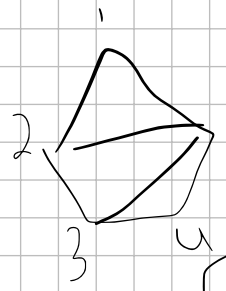
$n=5$

$X_{24} = X_{14} - X_{13}$

$X_{25} = -X_{13}$

X_{35}

→ does not involve X_{35}



X_{25}

X_{13}

X_{14}

X_{24}

factor $\dfrac{1}{X_{35}}$

$$\underbrace{\frac{a_{35\,55}}{X_{35}^2}}_{=0} + \frac{1}{X_{35}}\left(\boxed{\sum_{i\neq 35} \frac{a_{35\,ii}}{X_{ii}}}\right) + \sum_{i,j,k,l\neq 35} \frac{a_{ij\,kl}}{X_{ij}X_{kl}} = 0 \qquad \frac{0}{X_{35}}$$

$$A = \frac{a_{35\,13}}{X_{13}} + \frac{a_{35\,14}}{X_{14}} + \frac{a_{35\,24}}{X_{24}} + \frac{a_{35\,25}}{X_{25}} = 0 \quad \text{when} \quad \begin{cases} X_{24} = X_{14} - X_{13} \\ X_{25} = -X_{13} \end{cases}$$

try to use the constraints to kill two things.

send $X_{13} \to \infty$ $(X_{23} \to \infty)$ chose X_{14} arbitrary $(X_{24} \to \infty)$

so $\boxed{a_{35\,14} = 0}$

do the same for X_{24} arbitrary. → $\boxed{a_{35\,24}} = 0$

$$A = \frac{a_{35\,13}}{X_{13}} + \frac{a_{35\,25}}{X_{25}} = 0 \quad \text{when} \quad \{X_{25} = -X_{13}$$

← the $n=4$ case.

↳ $\boxed{a_{35\,13} = a_{35\,25}}$

$a_{35\,13} = a_{35\,25}$

$\boxed{a_{35\,14} = a_{35\,24} = a_{35\,35} = 0}$

→ shift 5 times fixes everything

$\boxed{n=6}$

$X_{24} = X_{14} - X_{13}$

$X_{25} = X_{15} - X_{13}$

$X_{26} = -X_{13}$

X_{35} X_{36}

X_{46}

(A)

$X_{13} = X_{63} - X_{62}$

$X_{14} = X_{64} - X_{62}$

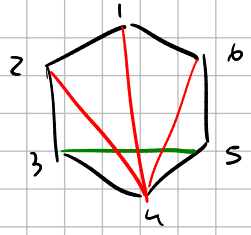
$X_{15} = -X_{62}$

X_{24} X_{25}

X_{35}

(B)

$$\frac{1}{X_{35}^3} \to 0$$



$$\frac{1}{X_{35}^2}\left(\frac{a_{(35)^2 13}}{X_{13}} + \frac{a_{(35)^2 14}}{X_{14}} + \frac{a_{(35)^2 15}}{X_{15}} + \frac{a_{(35)^2 24}}{X_{24}} + \frac{a_{(35)^2 25}}{X_{25}} + \frac{a_{(35)^2 26}}{X_{26}} + \frac{a_{(35)^2 36}}{X_{36}} + \frac{a_{(35)^2 46}}{X_{46}}\right) = 0$$

when (A) or (B) holds

(A)
- 46 is free $\to$ $a_{(35)^2 46} = 0$
- 36 is free $\to$ $a_{(35)^2 36} = 0$
- 13 fixed $(26 = -13)$ not $\to \infty$

  $a_{(35)^2 13} = a_{(35)^2 26}$
- $13 \to \infty$ chose 14, 15 arbitrary

  $a_{(35)^2 14} = a_{(35)^2 15} = 0$
- $13 \to \infty$ chose 24, 25 arbitrary

  $a_{(35)^2 24} = a_{(35)^2 25} = 0$

$X_{24} = X_{14} - X_{13}$ $\quad X_{35}$ X_{36}

$X_{25} = X_{15} - X_{13}$ $\quad X_{46}$

$X_{26} = -X_{13}$

(B) some steps

$\to$ everything is zero, and $15 = 62$

$a_{(35)^2 13} = a_{(35)^2 26} = 0$

$X_{13} = X_{63} - X_{62}$ $\quad X_{24}$ X_{25}

$X_{14} = X_{64} - X_{62}$ $\quad X_{35}$

$X_{15} = -X_{62}$

from this step we know that:

for $X_{i\,i+2}$ the cubed and squared terms have to be zero.

for $X_{i\,i+3}$ this is not true:

ex: X_{36} only (A) keeps it free: $\to$ $a_{(36)^2 13} = a_{(36)^2 26} = ?$

$$\frac{1}{X_{35}}\left(\frac{}{X_{14}^2} + \frac{1}{X_{14}X_{15}} + \frac{1}{X\ X} + \frac{1}{X\ X} + \frac{1}{X\ X} + \cdots\right.$$

$$\frac{1}{X_{13}^2} + \frac{1}{X_{14}^2} + \frac{1}{X_{13}X_{14}} + \frac{1}{X_{15}^2} + \frac{1}{X_{13}X_{15}} + \frac{1}{X_{14}X_{15}} + \frac{1}{X_{24}^2} + \frac{1}{X_{13}X_{24}} + \frac{1}{X_{14}X_{24}} +$$

$$\frac{1}{X_{15}X_{24}} + \frac{1}{X_{25}^2} + \frac{1}{X_{13}X_{25}} + \frac{1}{X_{14}X_{25}} + \frac{1}{X_{15}X_{25}} + \frac{1}{X_{24}X_{25}} + \frac{1}{X_{26}^2} + \frac{1}{X_{13}X_{26}} + \frac{1}{X_{14}X_{26}} +$$

$$\frac{1}{X_{15}X_{26}} + \frac{1}{X_{24}X_{26}} + \frac{1}{X_{25}X_{26}} + \frac{1}{X_{36}^2} + \frac{1}{X_{13}X_{36}} + \frac{1}{X_{14}X_{36}} + \frac{1}{X_{15}X_{36}} + \frac{1}{X_{24}X_{36}} + \frac{1}{X_{25}X_{36}} +$$

$$\frac{1}{X_{26}X_{36}} + \frac{1}{X_{37}^2} + \frac{1}{X_{13}X_{37}} + \frac{1}{X_{14}X_{37}} + \frac{1}{X_{15}X_{37}} + \frac{1}{X_{24}X_{37}} + \frac{1}{X_{25}X_{37}} + \frac{1}{X_{26}X_{37}} + \frac{1}{X_{36}X_{37}} +$$

$$\frac{1}{X_{46}^2} + \frac{1}{X_{13}X_{46}} + \frac{1}{X_{14}X_{46}} + \frac{1}{X_{15}X_{46}} + \frac{1}{X_{24}X_{46}} + \frac{1}{X_{25}X_{46}} + \frac{1}{X_{26}X_{46}} + \frac{1}{X_{36}X_{46}} + \frac{1}{X_{37}X_{46}}$$

45 terms

($n=5$ has 15 terms)

Z_{13}

$X_{24} = X_{14} - X_{13}$

$X_{25} = X_{15} - X_{13}$

$X_{26} = -X_{13}$

X_{35} X_{36} X_{46}

Z_{26}

$X_{13} = X_{63} - X_{62}$

$X_{14} = X_{64} - X_{62}$

$X_{15} = -X_{62}$

X_{24} X_{25}

X_{35}

we can send $13, 26 \to \infty$

close 36, 46 arbitrary and one each from (14, 24) (13, 25) arbitrary

(36 46 14 15) (36 46 24 15)

(36 46 14 25) (36 46 24 25)

---

same with Z_{26}

(24 25 13 14) (24 25 13 46)

(24 25 36 14) (24 25 36 46)

$$\frac{1}{X_{13}^2} + \frac{1}{X_{14}^2} + \frac{1}{X_{13}X_{14}} + \frac{1}{X_{15}^2} + \frac{1}{X_{13}X_{15}} + \frac{1}{X_{14}X_{15}} + \frac{1}{X_{24}^2} + \frac{1}{X_{13}X_{24}} + \frac{1}{X_{14}X_{24}} +$$

$$\frac{1}{X_{15}X_{24}} + \frac{1}{X_{25}^2} + \frac{1}{X_{13}X_{25}} + \frac{1}{X_{14}X_{25}} + \frac{1}{X_{15}X_{25}} + \frac{1}{X_{24}X_{25}} + \frac{1}{X_{26}^2} + \frac{1}{X_{13}X_{26}} + \frac{1}{X_{14}X_{26}} +$$

$$\frac{1}{X_{15}X_{26}} + \frac{1}{X_{24}X_{26}} + \frac{1}{X_{25}X_{26}} + \frac{1}{X_{36}^2} + \frac{1}{X_{13}X_{36}} + \frac{1}{X_{14}X_{36}} + \frac{1}{X_{15}X_{36}} + \frac{1}{X_{24}X_{36}} + \frac{1}{X_{25}X_{36}} +$$

$$\frac{1}{X_{26}X_{36}} + \frac{1}{X_{37}^2} + \frac{1}{X_{13}X_{37}} + \frac{1}{X_{14}X_{37}} + \frac{1}{X_{15}X_{37}} + \frac{1}{X_{24}X_{37}} + \frac{1}{X_{25}X_{37}} + \frac{1}{X_{26}X_{37}} + \frac{1}{X_{36}X_{37}} +$$

$$\frac{1}{X_{46}^2} + \frac{1}{X_{13}X_{46}} + \frac{1}{X_{14}X_{46}} + \frac{1}{X_{15}X_{46}} + \frac{1}{X_{24}X_{46}} + \frac{1}{X_{25}X_{46}} + \frac{1}{X_{26}X_{46}} + \frac{1}{X_{36}X_{46}} + \frac{1}{X_{37}X_{46}}$$

$\boxed{10 \text{ terms}}$

X_{26} squared survivor

$X_{13}X_{26}$ part of $X_{26} = -X_{13}$

$X_{15}X_{26}$ part of $X_{15} = -X_{62}$

---

z_{15}

$X_{62} = X_{52} - X_{51}$ $\quad X_{13} \quad X_{14}$

$X_{63} = X_{53} - X_{51}$ $\quad X_{24}$

$X_{64} = -X_{51}$

$\boxed{(24 \quad 13 \quad 14 \quad 62 \quad \cancel{35})}$

$\boxed{(24 \quad 13 \quad 14 \quad 25 \quad \cancel{35})}$

$$\frac{1}{X_{13}^2} + \frac{1}{X_{14}^2} + \frac{1}{X_{13}X_{14}} + \frac{1}{X_{15}^2} + \frac{1}{X_{13}X_{15}} + \frac{1}{X_{14}X_{15}} + \frac{1}{X_{24}^2} + \frac{1}{X_{13}X_{24}} + \frac{1}{X_{14}X_{24}} +$$

$$\frac{1}{X_{15}X_{24}} + \frac{1}{X_{25}^2} + \frac{1}{X_{13}X_{25}} + \frac{1}{X_{14}X_{25}} + \frac{1}{X_{15}X_{25}} + \frac{1}{X_{24}X_{25}} + \frac{1}{X_{26}^2} + \frac{1}{X_{13}X_{26}} + \frac{1}{X_{14}X_{26}} +$$

$$\frac{1}{X_{15}X_{26}} + \frac{1}{X_{24}X_{26}} + \frac{1}{X_{25}X_{26}} + \frac{1}{X_{36}^2} + \frac{1}{X_{13}X_{36}} + \frac{1}{X_{14}X_{36}} + \frac{1}{X_{15}X_{36}} + \frac{1}{X_{24}X_{36}} + \frac{1}{X_{25}X_{36}} +$$

$$\frac{1}{X_{26}X_{36}} + \frac{1}{X_{37}} + \frac{1}{X_{13}X_{37}} + \frac{1}{X_{14}X_{37}} + \frac{1}{X_{15}X_{37}} + \frac{1}{X_{24}X_{37}} + \frac{1}{X_{25}X_{37}} + \frac{1}{X_{26}X_{37}} + \frac{1}{X_{36}X_{37}} +$$

$$\frac{1}{X_{46}^2} + \frac{1}{X_{13}X_{46}} + \frac{1}{X_{14}X_{46}} + \frac{1}{X_{15}X_{46}} + \frac{1}{X_{24}X_{46}} + \frac{1}{X_{25}X_{46}} + \frac{1}{X_{26}X_{46}} + \frac{1}{X_{36}X_{46}} + \frac{1}{X_{37}X_{46}}$$

$X_{62} = X_{52} - X_{51}$ $\quad X_{13} \quad X_{14}$ → pos. of $(i \; i+3)$ chords

$X_{63} = X_{53} - X_{51}$ $\quad X_{24}$

$X_{64} = -X_{51}$

→ these are the positions of short chords $(i \; i+2)$

we can rotate $n-2$ times so that X_{35} end up in all orange spots other than the $X_{..} = -X_{..}$ eq.

↳ (all the $(i \; i+2)$ spots)

Z_{24}

$X_{35} = X_{25} - X_{24}$ $\qquad X_{46}$ X_{41}

$X_{36} = X_{26} - X_{24}$ $\qquad X_{51}$

$X_{31} = -X_{24}$

zero sets that contain X_{35} as free:

$(46\ 14\ 15\ \cancel{35}\ 36)$ $\qquad (46\ 14\ 15\ \cancel{35}\ 26)$

$$\frac{1}{X_{13}^2} + \frac{1}{X_{14}^2} + \frac{1}{X_{13}X_{14}} + \frac{1}{X_{15}^2} + \frac{1}{X_{13}X_{15}} + \frac{1}{X_{14}X_{15}} + \frac{1}{X_{24}^2} + \frac{1}{X_{13}X_{24}} + \frac{1}{X_{14}X_{24}} +$$

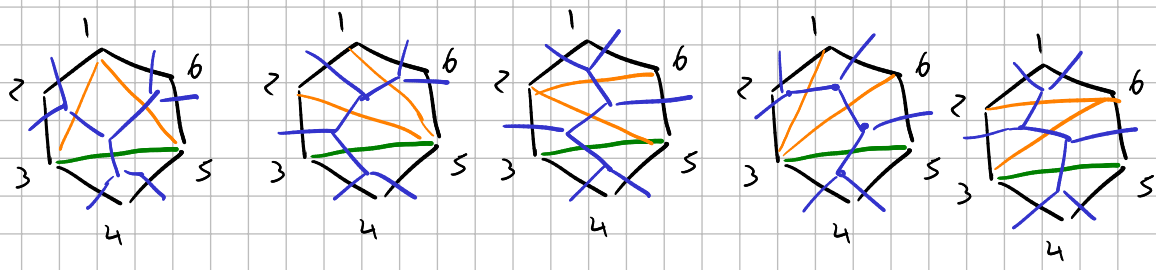
$$\frac{1}{X_{15}X_{24}} + \frac{1}{X_{25}^2} + \frac{1}{X_{13}X_{25}} + \frac{1}{X_{14}X_{25}} + \frac{1}{X_{15}X_{25}} + \frac{1}{X_{24}X_{25}} + \frac{1}{X_{26}^2} + \frac{1}{X_{13}X_{26}} + \frac{1}{X_{14}X_{26}} +$$

$$\frac{1}{X_{15}X_{26}} + \frac{1}{X_{24}X_{26}} + \frac{1}{X_{25}X_{26}} + \frac{1}{X_{36}^2} + \frac{1}{X_{13}X_{36}} + \frac{1}{X_{14}X_{36}} + \frac{1}{X_{15}X_{36}} + \frac{1}{X_{24}X_{36}} + \frac{1}{X_{25}X_{36}} +$$

$$\frac{1}{X_{26}X_{36}} + \frac{1}{X_{37}^2} + \frac{1}{X_{13}X_{37}} + \frac{1}{X_{14}X_{37}} + \frac{1}{X_{15}X_{37}} + \frac{1}{X_{24}X_{37}} + \frac{1}{X_{25}X_{37}} + \frac{1}{X_{26}X_{37}} + \frac{1}{X_{36}X_{37}} +$$

$$\frac{1}{X_{46}^2} + \frac{1}{X_{13}X_{46}} + \frac{1}{X_{14}X_{46}} + \frac{1}{X_{15}X_{46}} + \frac{1}{X_{24}X_{46}} + \frac{1}{X_{25}X_{46}} + \frac{1}{X_{26}X_{46}} + \frac{1}{X_{36}X_{46}} + \frac{1}{X_{37}X_{46}}$$

$\boxed{5 \text{ things}}$



all local $\qquad$ 13 35 15 $\qquad$ 35 25 26 $\qquad$ 35 25 26 $\qquad$ 35 13 16 $\qquad$ 35 36 26

13 35 51

for linear term : nilled 40 $\qquad$ with 35 fixed

for quadratic " : nilled 8

for cubic " : nill 1

what if factor out? $\boxed{36}$

try 7, 8, 9? with Mathematica